(Please write in CAPITAL LETTERS)

B

✓ Use the back **part** of the answer script for rough work. **No washroom breaks.**

✓ At the end of the exam, put the question **paper** inside the answer script and **return both.**

Question - 1: CO4 [6 Points]

Design the **ProbationAdvisor** class derived from the **Advisor** class so that the following driver code produces the given output.

- A probation advisor can advise as many students as they want
- Probation status: 0 < CGPA < 2

Parent Class & Driver Code

```
public class Advisor{
  public String name;
  public String type;
  public static int count;
  public Advisor(String name, String type){
    this.name = name;
    this.type = type;
  }
  public void advising(){
    System.out.println("Advising will start soon!");
  }
  public String toString(){
    return "Advisor Name: "+name+"\nAdvisor Type:
"+type;
  }
}
public class Tester{
 public static void main(String args []){
  System.out.println("Total Advisors: "+
Advisor.count);
  ProbationAdvisor p1 = new ProbationAdvisor("Saad");
  ProbationAdvisor p2 = new ProbationAdvisor("Reza");
  System.out.println("1----------------");
  System.out.println(p1);
  System.out.println("2----------------");
  p1.advising("Mithila", 3.5, true);
  p1.advising("Mou", 1.5, true);
  p1.advising("Afra", 1.0, false);
  p1.advising("Zarin", 2.7, false);
  p1.advising("Anika", 1.7, false);
  System.out.println("3----------------");
  System.out.println(p1);
  System.out.println("4----------------");
```

Output

```
Total Advisors: 0
1----------------
Advisor Name: Saad
Advisor Type: Probation
Total Advisee: 0
List:
2----------------
Mithila is not on probation
Please unlock the profile
first
Advising is complete for
Afra
Zarin is not on probation
Advising is complete for
Anika
3----------------
Advisor Name: Saad
Advisor Type: Probation
Total Advisee: 2
List: Afra Anika
4----------------
Advising is complete for Mim
Maliha is not on probation
5----------------
Advisor Name: Reza
Advisor Type: Probation
Total Advisee: 1
List: Mim
6----------------
All advisee info is cleared
for Saad
7----------------
Advisor Name: Saad
```

```
    p2.advising("Mim", 1.2, false);
    p2.advising("Maliha", 3.9, true);
    System.out.println("5-----------------");
    System.out.println(p2);
    System.out.println("6-----------------");
    p1.clearAllAdviseeInfo();
    System.out.println("7-----------------");
    System.out.println(p1);
    System.out.println("8-----------------");
    System.out.println("Total Advisors: "+
Advisor.count);
  }
}
```

```
Advisor Type: Probation
Total Advisee: 0
List:
8-----------------
Total Advisors: 2
```

Question - 2: CO2 [6 Points]

```
public class Food{
  String time = "everyday";
  public void serve() {
    if (this instanceof Food){
      System.out.println("Food is ready");
      System.out.println(time);
    } else if (this instanceof Snack){
      System.out.println(this);
      System.out.println((((Snack)this).time));
    }
  }
  public String toString(){

    return "Eating the food";
  }
}
public class Snack extends Food{
  String time = "now";
  public void waiting(){
    System.out.println(this + " " + time);
  }
}
```

```
public class Meal extends Food{
 public void waiting(){
  System.out.println("Waiting for meal");
  System.out.println((Meal)this);
 }
 public String toString() {
  return "Having a meal";
 }
}
public class Lunch extends Meal{
 public void serve(){
  System.out.println(super.toString()+"
"+time);
 }
}
public class BreakFast extends Meal{
 String time = "In the morning";
 public void waiting(){
  super.waiting();
  System.out.println(time+" "+ super.time);
 }
}
```

Assuming the following variables have been defined in the main method:

```
Snack obj1 = new Snack();
Food obj2 = new BreakFast();
Object obj3 = new Snack();
Food obj4 = new Lunch();
```

In the table to the right,

- Simulate the execution of each statement individually and find the outputs.
- **Write the outputs serially in your script.**
- For multi-line outputs: use slashes (as in "a/b/c"). For errors: write "CT error" or "RT error" to indicate when the error would be detected.

1	((Food)obj4).serve();
2	((Snack)obj3).serve();
3	System.out.println((Food)obj2);
4	obj1.waiting();
5	((Meal)obj2).waiting();
6	System.out.println((Meal)obj3);

Question - 3: CO3 [12 Points]

Write the **Drone** and **DroneController** classes to produce the given output.
Note:

- Drone controllers can overall activate a maximum of 5 drones.
- A drone controller can fly one drone at a time.
- Drone cannot get activated if battery is less than 10%.
- Battery percentage and droneID of a drone must be **private**.

Driver Code

```
public class DroneTester {
   public static void main(String[] args) {
       Drone drone1 = new Drone("DRN001", 9);
       Drone drone2 = new Drone("DRN002", 85);
       System.out.println("1===============");
       DroneController dc1 = new DroneController();
       System.out.println("2===============");
       dc1.activateDrone(drone1);
       System.out.println("3===============");
       drone1.chargeDrone(50);
       System.out.println("4===============");
       dc1.activateDrone(drone1);
       System.out.println("5===============");
       System.out.println("Active drones: " +
DroneController.getActiveDrones());
       drone1.displayStatus();
       System.out.println("6===============");
       dc1.flyDrone(drone1);
       System.out.println("7===============");
       drone1.displayStatus();
       System.out.println("8===============");
       dc1.flyDrone(drone2);
       System.out.println("9===============");
       dc1.landDrone(drone1);
       drone1.displayStatus();
       System.out.println("10===============");
       dc1.flyDrone(drone2);
       dc1.activateDrone(drone2);
       dc1.flyDrone(drone2);
       System.out.println("11===============");
       System.out.println("Active drones: " +
DroneController.getActiveDrones());
       dc1.systemShutDown();
       System.out.println("Active drones after shutdown:
" + DroneController.getActiveDrones());
   }
}
```

Output

```
1===============
2===============
Not enough battery power to
activate the drone.
3===============
Charged to 59 %.
4===============
DRN001 activated by the
controller.
5===============
Active drones: [DRN001, null,
null, null, null]
Drone ID: DRN001
Battery %: 59
Active status: true
Flying status: false
6===============
DRN001 in flight.
7===============
Drone ID: DRN001
Battery %: 59
Active status: true
Flying status: true
8===============
A drone controller can fly one
drone at a time.
9===============
Drone ID: DRN001
Battery %: 59
Active status: true
Flying status: false
10===============
Cannot fly unless activated.
DRN002 activated by the
controller.
DRN002 in flight.
11===============
Active drones: [DRN001, DRN002,
null, null, null]
Active drones after shutdown:
[null, null, null, null, null]
```

Question - 4: CO2 [6 Points]

```
public class A{
  public static int temp = 3;
  public int x = 5, sum = 2, y = 3;
  public A(){
     temp = (--temp) + temp*2;
  }
  public A(int x){
     y = temp - x;
     methodA(temp, y);
  }
  public void methodA(int temp, int y){
     y = (this.temp++) + y + temp;
     x = B.x + sum + this.y;
     sum = ++sum + this.x + y;
     System.out.println(x + " " + y+ " " +sum);
  }
}
public class B extends A{
   public static int x = 5;
   public B(){
      x = temp + 4 ;
      super.temp -= 1;
   }
   public B(B b){
      super(1);
      sum = b.sum;
      y = b.x;
   }
   public void methodA(int m, int n){
      y = x + this.y - m;
      x = x + super.x + temp;
      sum = x + m + super.sum + n;
      System.out.println(x + " " + y+ " " +sum);
   }
   public void methodB(int a, int b){
      int y = 2;
      y = a + this.y + y;
      x = y + this.x + super.temp;
      methodA(temp, b);
      sum = x + y + this.sum;
      System.out.println(x + " " + y+ " " +sum);
   }
}
```

Illustrate the output of the following statements.
[Your outputs won't be accepted without the workings]

```
B b1 = new B();
new B();
B b2 = new B(b1);
b1.methodB(3, 2);
```

Output:

Out1	Out2	Out3
26	23	32
67	9	30
67	8	57